

CO1-AT1- ASSIGNMENT

NAME : GNANASURYA.P

REG NO: 192412235

SUBJECT: Machine Learning

SUBJECT CODE: ITA0612

1. A parking facility has three sections containing cars of different colors. Section A has 5 red, 3 blue, and 2 black cars; Section B has 4 red, 6 blue, and 5 black cars; and Section C has 7 red, 2 blue, and 6 black cars. A section is chosen at random, but Section B is randomly twice as likely to be selected as each of the other sections due to higher usage. A selected car from the chosen section turns out to be blue. What is the probability that the car came from Section C?

Answer:

Prior probabilities

$$P_A = 1/4$$

$$P_B = 1/2$$

$$P_C = 1/4$$

Probability of selecting a blue car from each section

$$P_{\text{Blue}_A} = 3/10$$

$$P_{\text{Blue}_B} = 6/15$$

$$P_{\text{Blue}_C} = 2/15$$

Total probability of selecting a blue car

$$P_{\text{Blue}} = (P_A * P_{\text{Blue}_A}) + (P_B * P_{\text{Blue}_B}) + (P_C * P_{\text{Blue}_C})$$

Bayes theorem

$$P_{C_{\text{given_Blue}}} = (P_C * P_{\text{Blue}_C}) / P_{\text{Blue}}$$

```
print("Probability that the blue car came from Section C =", round(P_C_given_Blue, 4))
```

Probability that the blue car came from Section C = 0.1081

2. Examine the performance of a machine-learning model using its training and validation loss values over multiple epochs: (5 Marks)

- 1. (a) Epoch 1: Training Loss = 2.8, Validation Loss = 2.4**
- 2. (b) Epoch 2: Training Loss = 2.5, Validation Loss = 2.2**
- 3. (c) Epoch 3: Training Loss = 2.2, Validation Loss = 2.0**
- 4. (d) Epoch 4: Training Loss = 2.0, Validation Loss = 2.1**
- 5. (e) Epoch 5: Training Loss = 1.8, Validation Loss = 2.3**
- 6. (f) Epoch 6: Training Loss = 1.6, Validation Loss = 2.6**
- 7. (g) Epoch 7: Training Loss = 1.5, Validation Loss = 2.9**
- 8. (h) Epoch 8: Training Loss = 1.3, Validation Loss = 3.2**
- 9. i) Identify the epoch at which the model begins to overfit.**
- 10. ii) Suggest two effective techniques to reduce overfitting in this model.**
- 11. iii) Discuss how overfitting affects the model's generalization ability on unseen data.**

Answer:

```
import matplotlib.pyplot as plt

epochs = [1,2,3,4,5,6,7,8]

training_loss = [2.8,2.5,2.2,2.0,1.8,1.6,1.5,1.3]

validation_loss = [2.4,2.2,2.0,2.1,2.3,2.6,2.9,3.2]

plt.plot(epochs, training_loss, marker='o', label="Training Loss")
plt.plot(epochs, validation_loss, marker='s', label="Validation Loss")

plt.title("Training vs Validation Loss")

plt.xlabel("Epoch")

plt.ylabel("Loss")

plt.legend()

plt.grid(True)

plt.show()

# Find overfitting point
for i in range(1, len(validation_loss)):

    if validation_loss[i] > validation_loss[i-1]:

        print("Model starts overfitting at Epoch", epochs[i])
```

```

break

print("\nTechniques to reduce overfitting:")

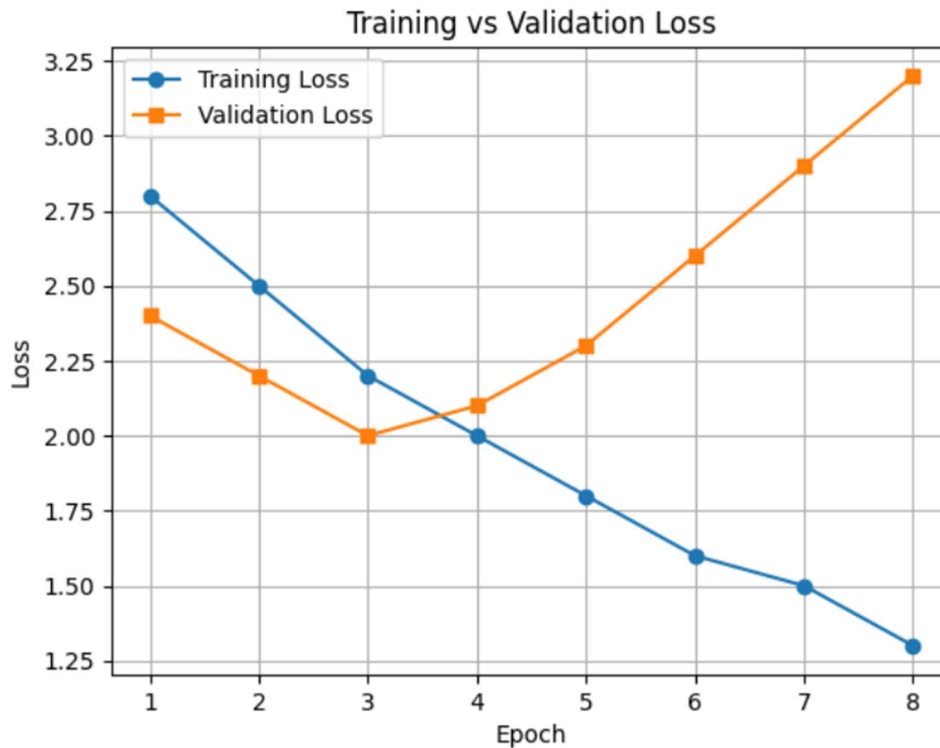
print("1. Early Stopping")

print("2. Dropout / Regularization")

print("\nEffect of Overfitting:")

print("The model memorizes training data and performs poorly on unseen data.")

```



3.Examine the performance of a machine-learning model using training and validation accuracy over multiple epochs: (5 Marks)

12. (a) Epoch 1: Training Accuracy = 60%, Validation Accuracy = 58%

13. (b) Epoch 2: Training Accuracy = 65%, Validation Accuracy = 63%

14. (c) Epoch 3: Training Accuracy = 70%, Validation Accuracy = 68%

15. (d) Epoch 4: Training Accuracy = 75%, Validation Accuracy = 72%

16. (e) Epoch 5: Training Accuracy = 80%, Validation Accuracy = 73%

17. (f) Epoch 6: Training Accuracy = 85%, Validation Accuracy = 72%

18. (g) Epoch 7: Training Accuracy = 88%, Validation Accuracy = 70%

19. (h)Epoch 8: Training Accuracy = 92%,Validation Accuracy = 68%

20. i) Identify the epoch at which the model starts overfitting.
21. ii) Explain why validation accuracy decreases despite training accuracy increasing. 22. iii) Suggest two methods to improve validation performance.

Answer:

```
import matplotlib.pyplot as plt

epochs = [1,2,3,4,5,6,7,8]
training_accuracy = [60,65,70,75,80,85,88,92]
validation_accuracy = [58,63,68,72,73,72,70,68]

plt.plot(epochs, training_accuracy, marker='o', label="Training Accuracy")
plt.plot(epochs, validation_accuracy, marker='s', label="Validation Accuracy")
plt.title("Training vs Validation Accuracy")
plt.xlabel("Epoch")
plt.ylabel("Accuracy (%)")
plt.legend()
plt.grid(True)
plt.show()

# Find overfitting point
for i in range(1, len(validation_accuracy)):
    if validation_accuracy[i] < validation_accuracy[i-1]:
        print("Model starts overfitting at Epoch", epochs[i])
        break

print("\nReason for decrease in validation accuracy:")
print("The model learns the training data too closely and fails to generalize.")
print("\nMethods to improve validation performance:")
print("1. Early Stopping")
print("2. Data Augmentation / Regularization")
```

